

Gate E0 report — run_v1.2_r1 (Paper IV v1.2)

Verdict: **PASS**. Evidence directory: 20_EVIDENCE/E0/.

E0 Identity — initial pass (2026-09-30, ~12:25 UTC)

Scripts (auditor-written): e0_identity.py, e0_source_consistency.py, e0_manifests.py. Raw outputs: E0_IDENTITY.json, E0_SOURCE_CONSISTENCY.json, E0_MANIFESTS.json, source_file_hashes.json, import_graph.json. Logs in 10_LOGS/E0_*.log.

Readiness

1. Owner requested execution in chat on 2026-09-30 → met. (AUDIT_TARGET_v1.2.json still says execution_authorized: false; that field predates the request and is not modified by the auditor.)
2. Target status READY_FOR_EXTERNAL_AUDIT; all mandatory bindings present; target sha256 d1eb542d...fdc29 equals the value in PREPARATION.json. 3–5. Bound author validation / internal audit exist by hash (contents deferred per mandate §3). Readiness is not an E0 PASS.

Source identity (piv-v12-fb459343d234)

- All 30+ bound files: sha256 and byte counts match (all_bound_hashes_match: true).
- Source ZIP: 615 members, no CRC errors, no unsafe paths/duplicates/symlinks.
- ZIP ↔ directory ↔ SOURCE_MANIFEST: 615 = 615 = 615, zero content mismatches.
- 607 .lean modules (recomputed); 8 non-Lean entries (FREEZE_SCOPE.json, docs/PUBLICATION_AUDIT.md, lakefile.toml, lake-manifest.json, lean-toolchain, 3 tools/*.py).
- FREEZE_SCOPE lists exactly the 19 targets named in the mandate; all exist as modules.
- Toolchain leanprover/lean4:v4.28.0; Mathlib rev 8f9d9cff6bd728b17a24e163c9402775d9e6a365. All 9 packages in the shared cache have git HEAD equal to the frozen manifest revs with 0 tracked modifications. Packages root is not a junction (realpath = itself).
- Import graph: union of the 19 target closures = 607 own modules (all modules), matching §7's statement that every local module lies in the closure of the 19 targets. **CORRECTION C-01 (auditor):** the first run of e0_source_consistency.py used an import regex that missed Lean module-system public import lines (21 occurrences, 20 files with module header) and reported 594 with 13 RequestProject.AFKS/AS modules outside every closure. That result was an auditor parsing error; the regex was fixed and rerun (10_LOGS/E0_source_consistency_run2.log). Original run log preserved (10_LOGS/E0_source_consistency.log). Static closure is still to be confirmed by E4's actual build graph.
- Lakefile declares lean_lib names PaperV, E18, E19 etc.; no PaperV.* module exists in the cut (OBSERVATION E0-O2, configuration residue, no content).
- Textual scan for sorry/admit/axiom/unsafe/implemented_by/extern/native_decide/opaque/ofReduceBool: all hits are inside doc comments (/ -! ... -/); no code-level occurrence. Kernel-level confirmation deferred to E4.

Manuscript identity (piv-v12-manuscripts-2451d43bface)

- MANUSCRIPT_MANIFEST (105 files) matches directory bytes; ZIP has 106 members = 105 + manifest, all hashes equal.
- Directory holds 212 additional unbound files (editorial reports, QA PNGs, history) — not part of the identity.
- PDFs: EN 68 pages, ES 69 pages (pymupdf, independently counted) — match the stated counts.

Author-build evidence identity

- EVIDENCE_MANIFEST: 1238 entries; all present on disk with matching hashes; ZIP has 1239 members (1238 + manifest) and every manifest hash occurs in the ZIP. Contents not read yet (deferred to E8).

Annex

- Annex ZIP: 40 members = 38 BoundedCliqueGap/*.lean + README + SOURCE_MANIFEST.sha256; all

39 manifest entries match ZIP content.

- v1.0 base ZIP: 508 members, 504 `.lean`, no `BoundedCliqueGap` files; its lakefile/manifest/toolchain equal the bound `lean_v1.0_freeze` files; lakefile declares `BoundedCliqueGap`; package revs equal v1.2's.

Status: **E0 initial checks PASS; final E0 requires the end-of-run immutability recheck.** Status: **E0 initial checks PASS; final E0 requires the end-of-run immutability recheck.**

Final recheck (2026-09-30 22:04 UTC)

`e0_identity.py` rerun: all 31 bound files and all ZIP integrity results identical to intake; source ZIP ↔ `□` directory ↔ `□` manifest still 615/615 with no mismatch (`10_LOGS/E0_identity_final.log`, `E0_IDENTITY_initial.json` vs `E0_IDENTITY.json`). **E0 = PASS.**

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